

Michelle (Michal) Merav

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EDUCATION

Columbia University

New York, NY

MS in Biomedical Engineering, CUNY Pathways Fellowship; GPA: 3.91/4.0

Expected Dec 2026

Relevant Courses: Biomaterials, Biomedical Design, Tissue Engineering, Image Computing, Computational Modeling

Hunter College (CUNY)

New York, NY

BA in Honors Biochemistry, BA in Studio Art GPA: 3.85/4.0

May 2024

Courses: Biophysical Chemistry, Computational Drug Discovery

INDUSTRY EXPERIENCE

Stratum Biosciences

New York, NY

Laboratory Scientist I

Aug 2025–June 2026

- Perform SA- β -gal and ORAC assays on primary NHDF cells for anti-senescence and antioxidative drug efficacy.
- Conduct drug diffusion studies using Franz cells and human trunk skin to evaluate transdermal drug delivery.
- Use Cursor to support operations for proprietary AI-driven skin health platforms and the Skin Angels network.

Sequential Bio

New York, NY

Research Associate

Sep 2024 – Aug 2025

- Perform DNA extraction and 384-well qPCR on collected samples to analyze skin microbiome health.
- Collected microbiome samples and skin barrier measurements for clinical research studies in the B2B pipeline.

ACADEMIC RESEARCH

Columbia University

New York, NY

Graduate Researcher, Biomedical Engineering

Aug. 2025-Present

- Encapsulate primary T-cells in polymer-nanoparticle hydrogels for immune microenvironment modulation.
- Chemically modify hyaluronic acid and hypromellose polymers under argon atmosphere and perform liposome synthesis.
- Synthesize azide-liposomes for hydrogel integration and test physical properties using DLS, rheology, and zeta potential.

Weill Cornell Medicine, Hunter College

New York, NY

Cancer Research Assistant, Chemistry

Jan 2025 – May 2025

- Created a 3D model of the tumor ECM in breast cancer using collagen-I, MB231 cultured cells, and IO-nanocages.
- Synthesized iron oxide nanocages for their use in anti-cancer drug delivery to tumor sites and imaged using TEM.

Hunter College

New York, NY

Neuroscience Research Assistant, Biological Sciences

Jun 2022 – Sep 2023

- Cultured and treated SY5Y neuronal cells and harvested for immunofluorescence, western blotting, and ELISA.
- Conducted rat perfusion for 15 subjects. Ran immuno-staining and used FIJI to edit microscopy images for analysis.
- Awarded “Best Poster” award at Hunter College Undergraduate Research Conference for presentation of findings.

Marine Biological Laboratory, University of Chicago

Woods Hole, MA

Research Assistant

Jul 2023 – Aug 2023

- Handled squid and removed giant axons for microdissection using precision scissors and forceps.
- Utilized Differential Interference Contrast (DIC) microscopy to record real-time transport in the squid axoplasm.

CERTIFICATES & TECHNICAL SKILLS

Certificates: FDNY Certificate of Fitness for Non-Production Chemical Laboratories (C-14)

Skills: Cell/Tissue Culture, FIJI (ImageJ), Microscopy (Confocal, Brightfield/Fluorescent, DIC, TEM), Rodent Handling, qPCR